

Gene flux erodes the karyotype barrier

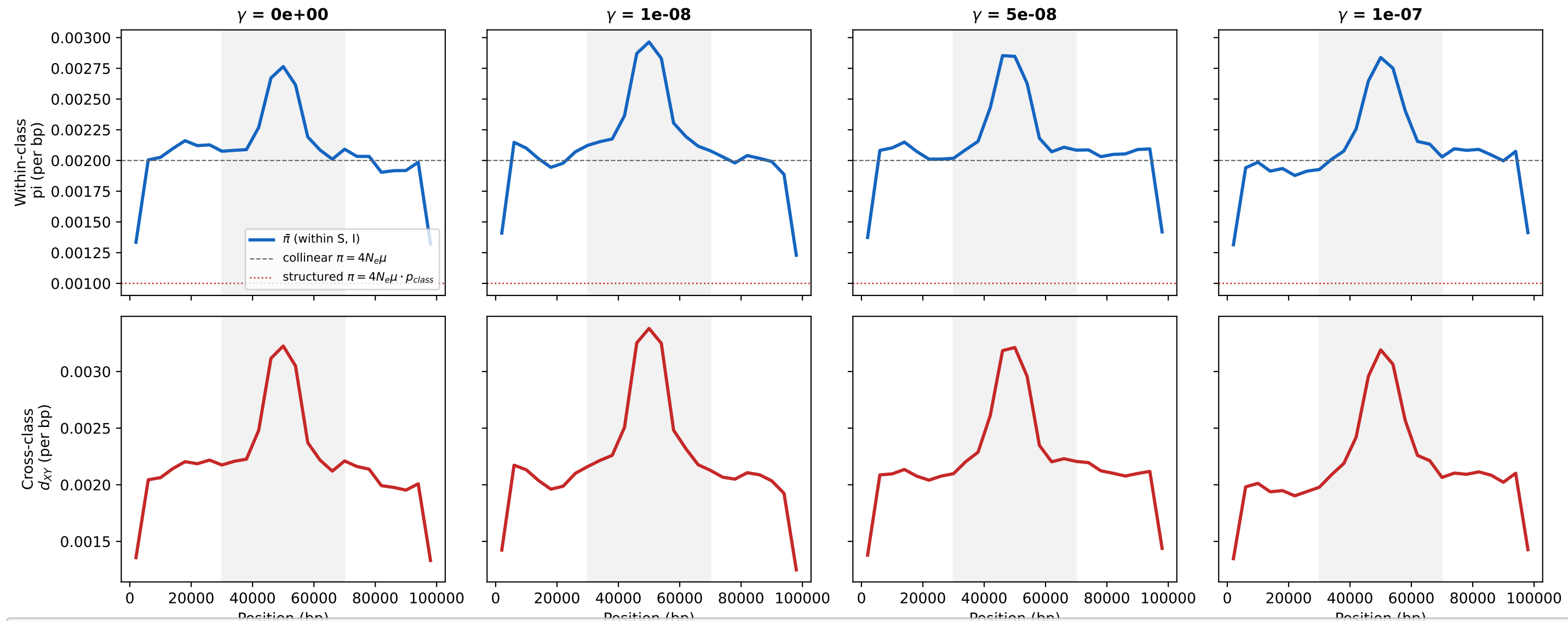


Figure. Effect of gene-conversion rate γ on the karyotype barrier. Each column shows a different γ value (0, $1e-8$, $5e-8$, $1e-7$ $\text{bp}^{-1} \text{gen}^{-1}$). (Top) Average within-class diversity $\bar{\pi} = (\pi_S + \pi_I)/2$. At $\gamma=0$ (no flux), $\bar{\pi}$ inside the inversion (grey shading) is depressed to $4N_e\mu \cdot p_{class}$ (red dashed) — the structured-coalescent prediction. As γ increases, gene flux relaxes the class barrier and $\bar{\pi}$ recovers toward the collinear expectation $4N_e\mu$ (grey dashed), first at the centre (where $\phi(x)$ peaks) and last at breakpoints. (Bottom) Cross-class d_{XY} (S vs I). At $\gamma=0$, d_{XY} is uniformly elevated; as γ grows, the centre collapses first, leaving breakpoints as the last barrier. Parameters: $N_e=50,000$, $p_{inv}=0.5$, $t_{inv}=200,000$ gen, $L=100$ kb, $\mu=1e-8$, $r=1e-8$, $n_S=n_I=8$, 40 replicates.

Command: `pixi run -e all python examples/gene_flux_demo.py`